

Combinators

Combinatory expressions

A *combinatory term* is an expression formed together by using standard combinators, functional application, and variables. We shall now define precisely what this means.

Let $\mathbb{V} = \{v_0, v_1, \dots\}$ be an infinite set of *variables*.

Let $\mathcal{C} = \{I, K, S, B, C, P, W, 1, c_2, \delta, \omega, \Omega\}$ be the set of *constants*.

The set $\mathbb{C}$ of combinatory terms, or combinatory expressions, is given by the grammar

$$\mathbb{C} ::= \mathbb{V} \mid \mathcal{C} \mid \mathbb{C}\mathbb{C}$$

We will use arbitrary lower-case letters $x, y, z, \dots$ to denote elements of $\mathbb{V}$.

The binary operation between combinators is known as *application*.

To reduce the number of parentheses, application is written with *association to the left*. For example, $ABCDE = (((AB)C)D)E$.

Notice that application is neither commutative nor associative, so $AB \neq BA$ and $A(BC) \neq (AB)C$. In particular, **the following kind of reasoning is invalid**:

$$\mathbf{X} \quad KII = K(II) = KI = K \quad \mathbf{Wrong!!}$$

This is wrong because neither the first step nor the last step are correct.

Carefully study the definitions given in the figure below.

These are the most famous combinators.

Identity	$Ix = x$
Konstant	$Kxy = x$
S-combinator	$Sxyz = xz(yz)$
Composition	$Bxyz = x(yz)$
exChange	$Cxyz = xzy$
Pairing	$Pxyz = zxy$
Duplication	$Wxy = xyy$
Application	$1xy = xy$
Numeral 2	$c_2xy = x(xy)$
Owl	$\delta xy = y(xy)$
Self-application	$\omega x = xx$
Omega	$\Omega = \omega\omega$

Figure 1: Standard Combinators

Reducing combinators

A *reduction* is a sequence of simplifications of a combinatory expression using the above rules from left to right.

In the following example, every time a rule is used, the corresponding subterm of the expression is underlined:

$$\begin{aligned} \text{WSI}(\text{C1})(\delta\text{I})(\text{KI}) &= \underline{\text{WSI}(\text{C1})}(\delta\text{I})(\text{KI}) \\ &= \underline{\text{SII}(\text{C1})}(\delta\text{I})(\text{KI}) \\ &= \underline{\text{I}(\text{C1})}(\underline{\text{I}(\text{C1})})(\delta\text{I})(\text{KI}) \\ &= \underline{\text{C1}(\text{C1})}(\delta\text{I})(\text{KI}) \\ &= \underline{1}(\delta\text{I})(\text{C1})(\text{KI}) \\ &= \underline{\delta\text{I}(\text{C1})}(\text{KI}) \\ &= \text{C1}(\underline{\text{I}(\text{C1})})(\text{KI}) \\ &= \underline{\text{C1}(\text{C1})}(\text{KI}) \\ &= \underline{1(\text{KI})(\text{C1})} \\ &= \underline{\text{KI}(\text{C1})} \\ &= \text{I} \end{aligned}$$

This reduction resulted in a term that cannot be reduced anymore; such an expression is called a *normal form*.

For example, $\text{K}x$, $y\text{I}$ are normal forms.

Sometimes, a reductions can end up in a *reduction cycle*:

$$\begin{aligned} \underline{\text{SII}(\text{SII})} &= \underline{\text{I}(\text{SII})}(\text{I}(\text{SII})) \\ &= \text{SII}(\underline{\text{I}(\text{SII})}) \\ &= \text{SII}(\text{SII}) \end{aligned}$$

In this case, the reduction ended in exactly the same term we started with, so it cannot be terminated. But sometimes, you will find a loop in a *subexpression* of the final term. This also signifies non-termination.

Practice problems

Simplify the following terms as much as possible, until either a normal form is reached, or an unescapable reduction cycle is produced.

1. $KK(II)$

2. $I IKKSS$

3. $SIK(KI)$

4. $K(K(KI))xyz$

5. $I(KI)I(KI)$

6. $I(KI)I(KI)I$

7. $SIK(SK)x$

8. $SSSIIIxyz$

9. $\omega(Bx\omega)$

10. $WSI(BK(SII))$

The answers to the above problems are given in a footnote on the next page.
Don't check them until you tried the problems yourself!

¹_{K,KS,I,I,KI,I,x,xyz,loop,loop}